

TECHNICAL REPORT

Global AQI Web Analysis

Benchmarking LSTM, CNN, and Transformer architectures for global air quality forecasting, with production deployment via Streamlit

Report Type: Final Project Report — Phases 1–3
Domain: Environmental Data Science / Deep Learning
Deployment Target: Streamlit Web Application

Executive Summary

This report documents the end-to-end development of a deep learning system for predicting global Air Quality Index (AQI) values from pollutant concentrations and geographic coordinates. The project progressed through three phases: data understanding and preprocessing, literature review and model selection, and model training, optimization, and deployment.

Three candidate architectures — LSTM, CNN, and Transformer — were implemented and benchmarked on an identical preprocessed dataset. The Transformer architecture achieved the strongest performance, with a test MSE of 32.87 and an R^2 score of 0.94, and was deployed as the production model in an interactive Streamlit application.

Key Results

Metric	Value
Best-performing model	Transformer
Test MSE	32.87
R^2 score	0.94
Models benchmarked	3 (LSTM, CNN, Transformer)
Deployment platform	Streamlit web application

1. Problem Statement

Air Quality Index (AQI) is a critical metric indicating air quality and associated public health risk. Predicting AQI from environmental parameters — including carbon monoxide (CO), nitrogen dioxide (NO2), particulate matter (PM2.5), ozone, and geographic location (latitude, longitude) — supports timely mitigation efforts and informed policy-making.

Objectives

- Build a deep learning model that accurately predicts city-level AQI from pollutant values and location.
- Deliver a deployable, interactive tool for real-time AQI prediction.

2. Dataset

The dataset consists of a global AQI CSV file containing pollutant readings paired with geographic coordinates.

Attribute	Detail
Source	Global AQI dataset (CSV) with latitude and longitude
Core fields	Country, City, AQI Value, AQI Category, CO/NO2/PM2.5/Ozone AQI values and categories, Latitude, Longitude
Shape	14 columns; row count per source file
Missing values	Mean imputation (numeric), mode imputation (categorical)

3. Exploratory Data Analysis

- AQI distribution: right-skewed, with most cities reporting moderate AQI levels.
- PM2.5: contains outliers; the majority of values fall in the moderate range.
- NO2, CO, and Ozone: predominantly moderate, with occasional high readings.
- Correlation: PM2.5 shows the strongest positive correlation with AQI; other pollutants correlate moderately.

Visualizations Produced

- Histogram of AQI value distribution.
- Boxplots of pollutant distributions.
- Correlation heatmap across all pollutant features.

4. Preprocessing Pipeline

- Missing numeric values imputed with column mean; missing categorical values imputed with column mode.
- Categorical features (Country, City) encoded using LabelEncoder.
- Numeric features standardized using StandardScaler.

Feature–Target Split

Role	Fields
Features	Country, City, Latitude, Longitude, CO AQI, NO2 AQI, PM2.5 AQI, Ozone AQI
Target	AQI Value

5. Literature Review and Model Selection

AQI prediction approaches in the literature span three principal architecture families: time-series models such as LSTM, spatial pattern models such as CNN, and attention-based models such as Transformers. Each was evaluated for suitability against the structure of the dataset.

Model	Strength	Limitation
LSTM	Captures sequential dependencies in pollutant trends	Slower to train; prone to overfitting
CNN	Efficient at learning local/spatial pollutant patterns	Limited capture of sequential information
Transformer	Captures long-range dependencies; flexible	Requires more data and compute

Given the sequential nature of pollutant trends, LSTM was selected as the baseline model, with CNN and Transformer trained as comparative architectures.

Implementation Plan

- Baseline architecture: LSTM (fully connected + sequential layers).
- Comparative architectures: CNN, Transformer.
- Train/test split: 80% / 20%.
- Hyperparameters tuned against a held-out validation set.

6. Model Training and Optimization

Training Pipeline

- All three architectures trained on an identical preprocessed dataset for direct comparability.
- EarlyStopping applied to prevent overfitting.
- Models evaluated using Mean Squared Error (MSE) and R² score.

Performance Comparison

Model	Test MSE	R ² Score
LSTM	34.21	0.92

Model	Test MSE	R ² Score
CNN	38.45	0.89
Transformer	32.87	0.94

Optimization and Hyperparameter Tuning

- Number of layers, units per layer, and learning rate adjusted iteratively.
- Dropout applied to reduce overfitting risk.
- Transformer architecture achieved the best generalization across the validation set.

7. Deployment and Results

Deployment

Component	Detail
Tool	Streamlit application
Input	Country, City, Latitude, Longitude, pollutant AQI values
Output	Predicted AQI value
Model artifacts	Trained model (.h5), label encoders (.pkl), feature scaler (.pkl)

Result Analysis

- Best model: Transformer, with the lowest test MSE and highest R² score of the three architectures.
- Final hyperparameters: 2 Transformer encoder layers, 64 units per layer, learning rate of 0.001.
- MSE indicates model predictions fall within a reasonable error margin for practical deployment.

In-App Visualizations

- Predicted-vs-actual AQI scatter plot.
- Feature importance derived from Transformer attention weights.

8. Conclusion

- Successfully built an end-to-end pipeline for global AQI prediction, from raw data to deployed application.
- Three deep learning architectures implemented, trained, and rigorously evaluated.
- The Transformer model outperformed both LSTM and CNN on test MSE and R² score.

- Final model deployed in an interactive Streamlit application for real-time, global AQI prediction.

Deliverables Summary

Phase	Deliverable	Submission Date
Phase 1	Report + Python notebook (EDA & preprocessing)	31 July 2025
Phase 2.1	Report + notebook with baseline & shortlisted models	Week 8
Phase 2.2	Notebook with complete pipeline, optimization & documentation	Week 9
Phase 3	Report + deployed Streamlit app + performance visualizations	Week 11